

Generative AI (CS F437)

Assignment 1

Assignment Title

Training and Evaluating Transformer Models for English to Hindi Translation

Objective

This assignment is designed to provide students with practical experience in training, fine-tuning, and evaluating transformer-based models for machine translation tasks. Students will utilize the IITB English-Hindi Parallel Corpus to implement transformer architectures, compare their results with trained models available on Hugging Face, and gain a comprehensive understanding of the following key aspects:

- Dataset preprocessing for neural machine translation (NMT).
- Training a transformer-based model from scratch for English-to-Hindi translation.
- Fine-tuning a pre-trained model to enhance translation performance.
- Evaluating and comparing different approaches using standard machine translation metrics.

Dataset: IITB English-Hindi Parallel Corpus

The IITB English-Hindi dataset is a high-quality parallel corpus developed by the Centre for Indian Language Technology (CFILT) at IIT Bombay. It is widely recognized for training and evaluating neural machine translation (NMT) models.

Dataset Features

- **Languages:** English (source) and Hindi (target).
- **Size:** Over 1.66 million parallel sentence pairs.
- **Format:** Available in Parquet format on Hugging Face.

Example Data

English Sentence	Hindi Translation
"Give your application an accessibility workout."	"अपने अनुप्रयोग को पहुंचनीयता व्यायाम का लाभ दें।"
"The default plugin layout for the bottom panel."	"निचले पैनल के लिए डिफॉल्ट प्लग-इन खाका।"
"Highlight duration."	"अवधि को हाइलाइट करें।"

Figure 1: Sample English-Hindi Parallel Corpus

Data Splits

- **Training Set:** 1.66M sentence pairs (rows).
- **Validation Set:** 520 sentence pairs (rows).
- **Test Set:** 2.51K sentence pairs (rows).

Applications of the Dataset

- **Neural Machine Translation (NMT):** Building AI models for English-to-Hindi translation.
- **Cross-Language Understanding:** Training multilingual AI systems.
- **AI-Powered Chatbots:** Developing conversational AI for Hindi-speaking users.

For more details, visit: [Hugging Face IITB English-Hindi Dataset](#).

Assignment Tasks

Part 1: Fine-Tuning a Transformer Model

In this section, students will fine tune a transformer model for English-to-Hindi translation.

- Choose a Transformer Architecture: BERT, T5, MarianMT, mBART, GPT, etc.
- Preprocess the Dataset:
 - Tokenize English and Hindi text.
 - Apply text normalization and padding.
 - Convert sentences into input tensors.
- Fine tune the chosen transformer Model on the IITB dataset.
- Evaluate the Model using BLEU, ROUGE, and METEOR scores.
- Document results including performance, challenges, and insights.

Part 2: Trained Transformer Model

In this section, students will use an existing trained transformer model to improve translation accuracy.

- Choose an existing trained model from Hugging Face, such as:
 - Helsinki-NLP/opus-mt-en-hi (MarianMT)
 - facebook/mbart-large-50 (mBART)
 - gemma-2b-mt-Hindi-Fine Tuned, etc
- Evaluate the chosen model using BLEU, ROUGE, METEOR scores.
- Compare part 2 results with the part 1 and document them including performance and insights.

Part 3: Comparative Analysis & Reporting

- Compare accuracy of both models.
- Analyze translation quality using evaluation metrics.
- Discuss pros/cons of training from scratch vs. fine-tuning.
- Conclude which approach is more efficient and suitable for real-world applications.

Deliverables

Code Implementation

- Python script (.py) or Jupyter Notebook (.ipynb) with comments.
- Structured using Hugging Face Transformers, PyTorch, or TensorFlow.
- Training and evaluation results logged and visualized.

Technical Report (4-6 pages, PDF format)

- **Introduction:** Overview of the dataset and problem statement.
- **Methodology:** Steps taken for fine-tuning and testing for both models.
- **Results & Comparison:**
 - Performance metrics for both models.
 - Training time, accuracy, etc.
- **Conclusion:** Insights on the best approach and its real-world applications.
- **References:** List of resources and research papers used.

Submission Guidelines

- **Deadline:** 27/03/2025
- **Submission Format:** ZIP file containing (id-name.zip):
 - Code files (.py or .ipynb)
 - Report (PDF format)
- **Submission Platform:** Google Classroom / LMS .

Additional Notes

- ✓ • Save trained models and submit them with code.
- ✓ • Use Google Colab or Kaggle for GPU utilization.
- Experiment with different hyperparameters.
- ✓ • Provide visualizations (loss curves, accuracy graphs, sample translations).
- Bonus points for low-resource translation challenges and improvements.

Good luck!